



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SHRI MANJUNATHA FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

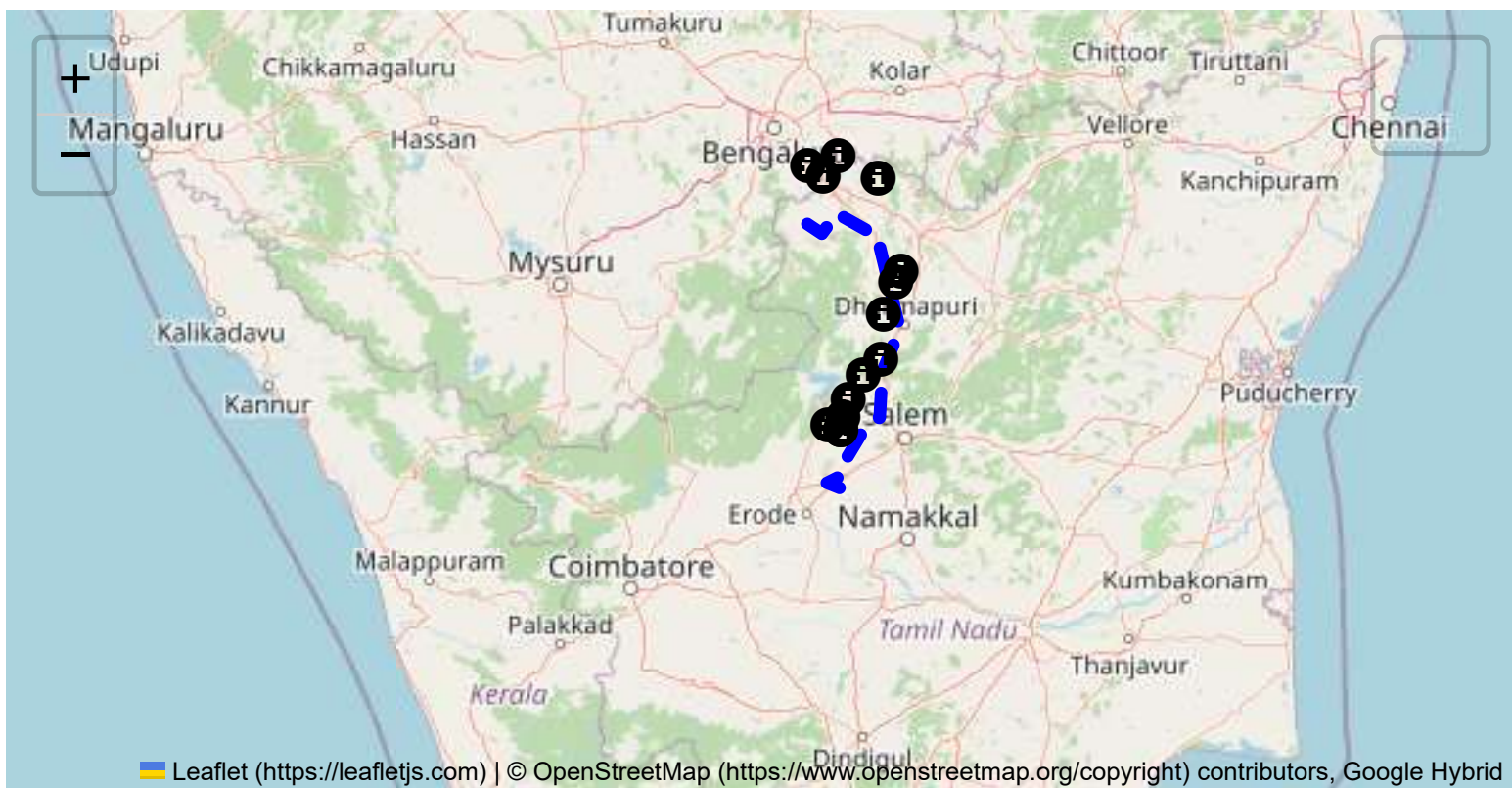
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

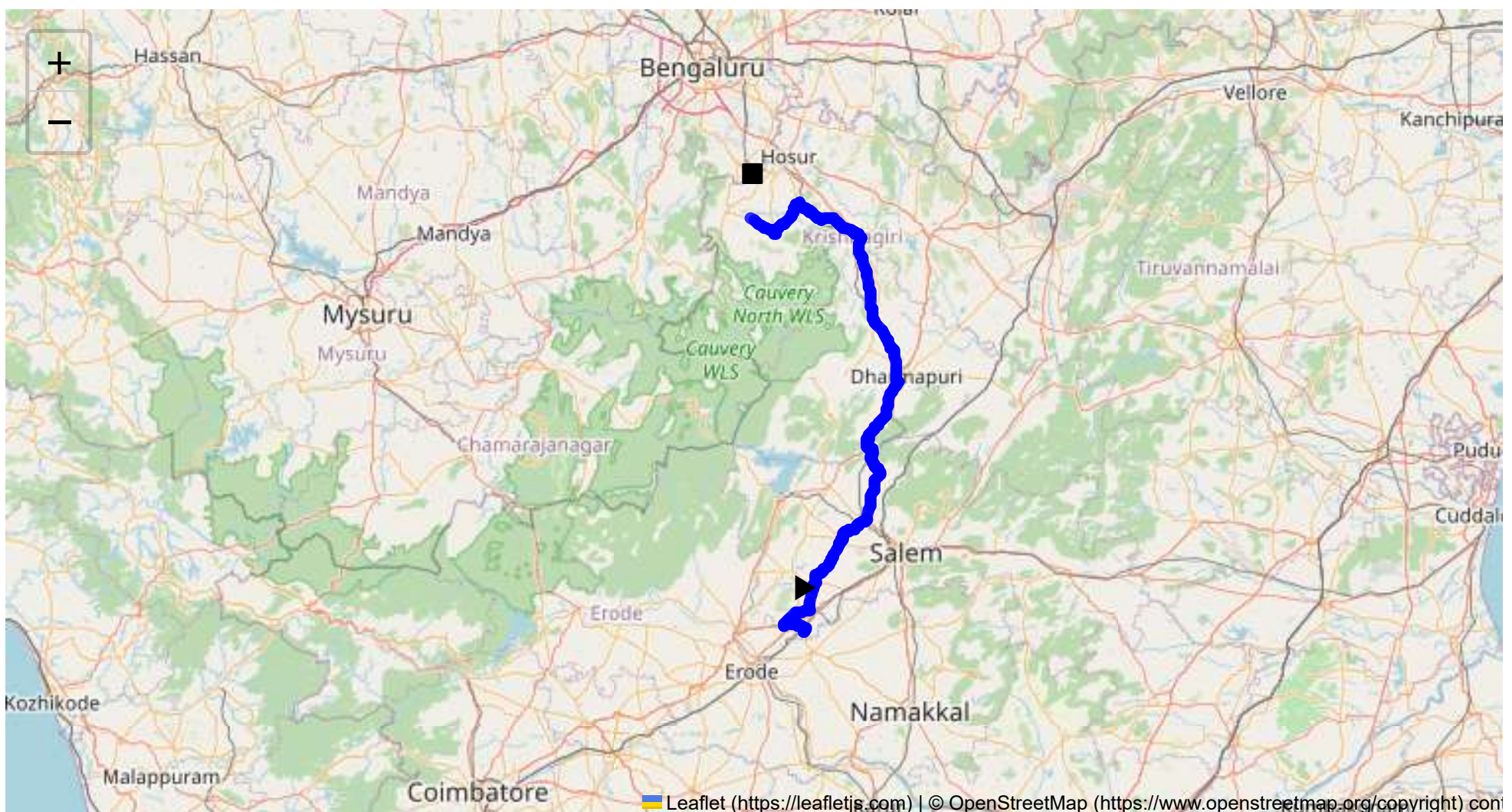
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 194.28 km
Estimated Duration: 4.1 hours
Adjusted Duration (Heavy Vehicle): 5.1 hours
Start: (11.4381, 77.8734)
End: (12.563323, 77.723501)



Welcome to the Journey Risk Management Study

Route Overview

This route begins at Sangagiri (CVQF+23W), passes through a series of towns and strategic points, and ends at Thogarai Agraharam (HP7F+6W), covering approximately 194.28 kilometers. The journey connects key highways and arterial roads such as the Salem-Kochi Highway and SH 86, serving as a crucial link for heavy vehicles.

Weather Conditions and Hazards

Tamil Nadu generally experiences a tropical climate. The route is subject to:

- **Heat Waves:** During the summer months (March to June).
- **Monsoons:** Heavy rains from June to September can cause localized flooding or water-logged roads.
- **Fog:** Possible in winters (December to February), particularly in early mornings, which may affect visibility.

Traffic Patterns

- **Peak Hours:** Typically, 8-10 AM and 5-7 PM, especially near Sangagiri, Salem, and Dharmapuri.
- **Congestion Prone Areas:** Salem urban region and junctions at major highways, such as Thoppur junction.

Road Quality and Infrastructure

- **Highways:** Generally well-maintained, but watch for construction zones and maintenance work, especially on smaller state highways.
- **Local Roads:** Varying quality with occasional potholes and narrow stretches which might require cautious driving.

Alternative Routes for Emergencies

- **Salem - Hosur Highway (NH 44):** An alternative major highway parallel to a significant section of the route.
- **State Highways:** Connecting through nearby small towns which may be used to bypass bottlenecks.

Regulations on Hazardous Material Transport

- Typically require special permits for hazardous materials.
- Restrictions on timing often apply, potentially prohibiting travel during peak hours or through highly populated areas.

Historical Incidents

- Occasional reports of overturned trucks due to sharp turns or potholes, particularly in hilly segments.
- Instances of congestion-related accidents, highlighting the need for vigilance in traffic.

Environmental and Sensitive Areas

- The region hosts several agricultural zones requiring careful navigation to avoid spilling hazardous materials.
- The route does not pass directly through major protected areas, but awareness of local flora and fauna should be maintained.

Communication Coverage

- Generally good across highway segments with localized dead zones in rural areas.
- Important to plan for these by maintaining alternative contact methods (satellite phones, radio communication).

Emergency Response Times

- Varies widely, typically longer in rural or less densely populated sections.
- Urban stretches, like those near Salem, have quicker response times due to proximity to emergency services.

Risk Assessment Summary

This route, while traversing several urban and rural interfaces, is feasible for heavy trucks transporting hazardous materials with specific considerations:

- Monitor weather conditions, particularly during monsoon.
- Avoid peak hours where possible to reduce the risk of congestion-related delays.
- Ensure vehicles comply with local transport regulations for hazardous materials.
- Maintain communication and emergency contacts readily accessible due to possible coverage gaps.

Overall, with strategic planning regarding timing and weather conditions, the route is manageable for professional drivers. However, awareness and caution are necessary for safe and efficient transport operations.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	12.60628, 77.86014	15 KM/Hr
1	Turn	High	11.43961, 77.87341	15 KM/Hr
2	Turn	High	11.44077, 77.87533	15 KM/Hr
3	Turn	High	11.44898, 77.87410	15 KM/Hr
4	Turn	Medium	11.45352, 77.85698	30 KM/Hr
5	Turn	Medium	11.45764, 77.85219	30 KM/Hr
6	Turn	High	11.46303, 77.84945	15 KM/Hr
7	Turn	Medium	11.45545, 77.81812	30 KM/Hr
8	Turn	High	11.45584, 77.81695	15 KM/Hr
9	Turn	Medium	11.45656, 77.81709	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
10	Turn	High	11.49383, 77.88442	15 KM/Hr
11	Turn	Medium	11.56352, 77.90078	30 KM/Hr
12	Turn	Medium	11.59189, 77.90645	30 KM/Hr
13	Turn	Medium	11.59328, 77.90809	30 KM/Hr
14	Turn	Medium	11.63867, 77.94841	30 KM/Hr
15	Turn	High	11.64217, 77.95200	15 KM/Hr
16	Turn	Medium	11.64712, 77.95118	30 KM/Hr
17	Turn	Medium	11.67858, 77.96834	30 KM/Hr
18	Turn	Medium	11.69776, 77.97983	30 KM/Hr
19	Turn	Medium	11.70179, 77.97900	30 KM/Hr
20	Turn	Medium	11.71394, 77.99572	30 KM/Hr
21	Turn	Medium	11.71916, 78.01117	30 KM/Hr
22	Turn	Medium	11.72504, 78.01680	30 KM/Hr
23	Turn	Medium	11.73766, 78.03445	30 KM/Hr
24	Turn	High	11.73651, 78.03868	15 KM/Hr
25	Turn	Medium	11.74120, 78.04588	30 KM/Hr
26	Turn	Medium	11.86805, 78.08353	30 KM/Hr
27	Turn	Medium	11.87913, 78.08206	30 KM/Hr
28	Turn	Medium	11.91159, 78.05857	30 KM/Hr
29	Turn	Medium	11.92566, 78.06503	30 KM/Hr
30	Turn	Medium	11.93307, 78.06426	30 KM/Hr
31	Turn	Medium	11.93545, 78.06130	30 KM/Hr
32	Turn	Medium	11.93694, 78.05425	30 KM/Hr
33	Turn	Medium	11.95648, 78.04768	30 KM/Hr
34	Turn	High	11.95803, 78.04964	15 KM/Hr
35	Turn	Medium	11.96068, 78.04960	30 KM/Hr
36	Turn	High	11.96370, 78.05142	15 KM/Hr
37	Turn	High	11.96650, 78.04927	15 KM/Hr
38	Turn	Medium	12.12094, 78.13269	30 KM/Hr
39	Turn	High	12.51318, 78.03193	15 KM/Hr
40	Turn	High	12.51402, 78.03107	15 KM/Hr
41	Turn	Medium	12.51369, 78.02985	30 KM/Hr
42	Turn	Medium	12.52184, 78.02409	30 KM/Hr
43	Turn	Medium	12.52732, 78.00940	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
44	Turn	Medium	12.52925, 78.00830	30 KM/Hr
45	Turn	Medium	12.54115, 77.98118	30 KM/Hr
46	Turn	High	12.55126, 77.97205	15 KM/Hr
47	Turn	Medium	12.55328, 77.96518	30 KM/Hr
48	Turn	Medium	12.56256, 77.92252	30 KM/Hr
49	Turn	Medium	12.58309, 77.89268	30 KM/Hr
50	Turn	Medium	12.58408, 77.88981	30 KM/Hr
51	Turn	Medium	12.58547, 77.88101	30 KM/Hr
52	Turn	Medium	12.58950, 77.87819	30 KM/Hr
53	Turn	Medium	12.59196, 77.86963	30 KM/Hr
54	Turn	Medium	12.59972, 77.86299	30 KM/Hr
55	Turn	Medium	12.60233, 77.86275	30 KM/Hr
56	Turn	Medium	12.60511, 77.85508	30 KM/Hr
57	Turn	Medium	12.60188, 77.85288	30 KM/Hr
58	Turn	Medium	12.60131, 77.85133	30 KM/Hr
59	Turn	Medium	12.54884, 77.81468	30 KM/Hr
60	Blind Spot	Blind Spot	12.52432, 77.79161	10 KM/Hr
61	Turn	High	12.52496, 77.79040	15 KM/Hr
62	Turn	High	12.53221, 77.78843	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
3	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
4	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
5	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
11	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
12	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
13	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
14	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
15	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
16	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
17	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
18	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
19	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
20	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
22	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
23	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
27	hospital	Mythri Hospital, Krishnagiri	12.5264111, 77.7897745	30 km/h	Medium
28	pharmacy	Sri Venkateshwara Pharmacy	12.5319785, 77.7884257	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
21	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
24	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
25	marketplace	Rayakottai tomato market	12.5237469, 78.0220377	30 km/h	Medium
26	school	St. Jopseph School	12.5339473, 77.795134	30 km/h	Medium
29	school	Vellankani Hr. Sec. School	12.5336092, 77.7899878	30 km/h	Medium
30	school	Play Ground	12.5324778, 77.7897856	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.60628, 77.86014



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44077, 77.87533



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45764, 77.85219



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45545, 77.81812



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45584, 77.81695



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45656, 77.81709



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49383, 77.88442



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56352, 77.90078



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59189, 77.90645



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59328, 77.90809



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63867, 77.94841



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64217, 77.95200



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64712, 77.95118



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67858, 77.96834



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69776, 77.97983



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.70179, 77.97900



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71394, 77.99572



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71916, 78.01117



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72504, 78.01680



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73766, 78.03445



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73651, 78.03868



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74120, 78.04588



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86805, 78.08353



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87913, 78.08206



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.91159, 78.05857



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.92566, 78.06503



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93307, 78.06426



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93545, 78.06130



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93694, 78.05425



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95648, 78.04768



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.95803, 78.04964



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96068, 78.04960



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96370, 78.05142



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96650, 78.04927



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.51318, 78.03193



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.51402, 78.03107



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.51369, 78.02985



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.52184, 78.02409



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.52732, 78.00940



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.52925, 78.00830



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.54115, 77.98118



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.55126, 77.97205



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.55328, 77.96518



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.56256, 77.92252



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.58309, 77.89268



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.58408, 77.88981



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.58547, 77.88101



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.58950, 77.87819



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.59196, 77.86963



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.59972, 77.86299



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.60233, 77.86275



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.60511, 77.85508



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.60188, 77.85288



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.60131, 77.85133

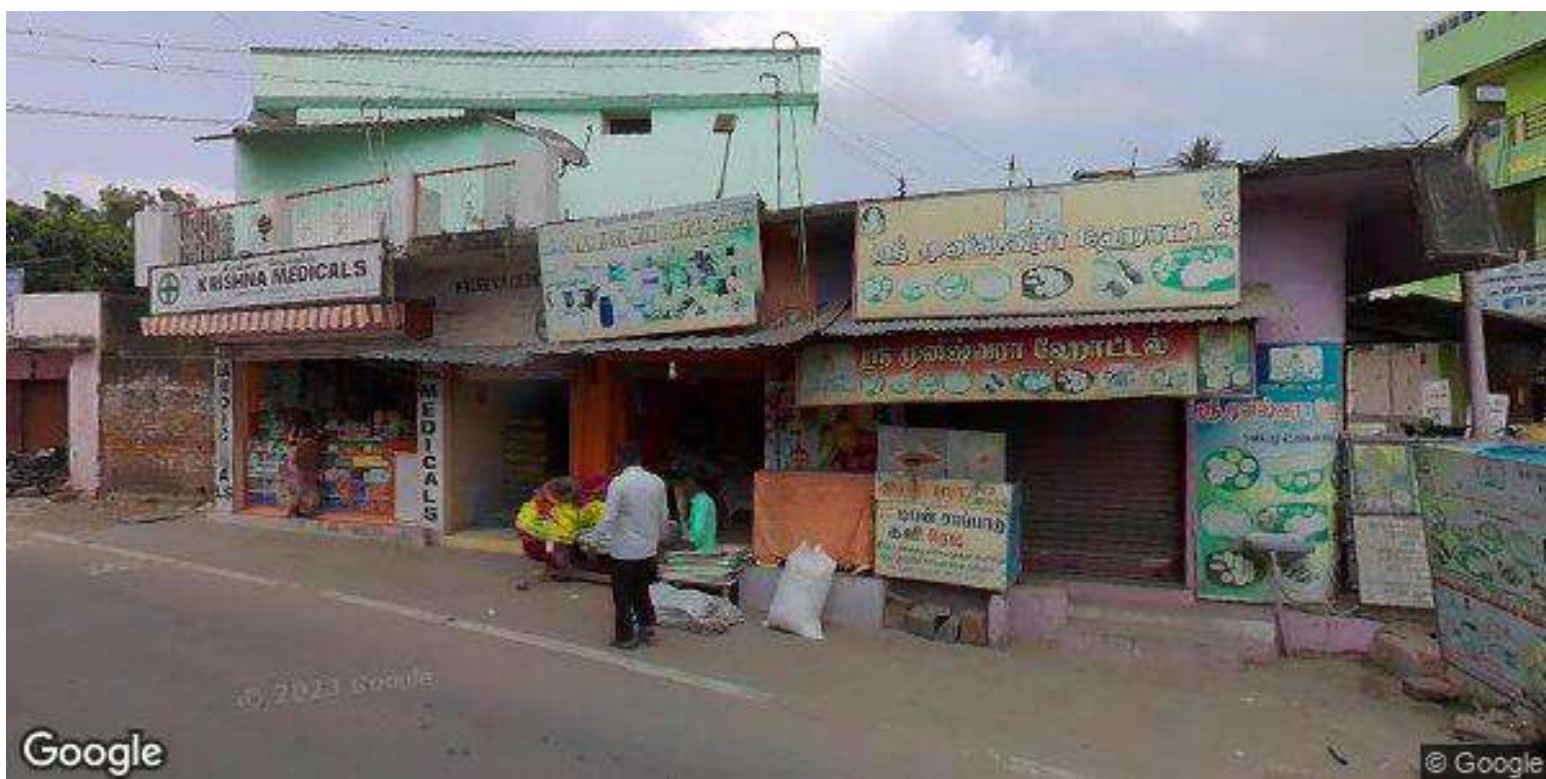


Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.54884, 77.81468



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 12.52432, 77.79161



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.52496, 77.79040



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.53221, 77.78843
